

Matrix functions and stability

Finiteness for nonnegative rational matrix families

Difficulty: extreme **Importance:** broadly interesting**Status:** Partially resolved

Rating rationale: Extreme because rational-input finiteness is a longstanding barrier in joint spectral-radius theory; its consequences reach control, combinatorics, and exact computation.

Context and notation

The joint spectral radius of a nonempty compact set $\mathcal{M} \subset \mathbb{C}^{d \times d}$ is

$$\hat{\rho}(\mathcal{M}) = \lim_{k \rightarrow \infty} \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

Problem statement

Does every finite nonempty set $\mathcal{M} \subset \mathbb{Q}_{\geq 0}^{d \times d}$, for every positive integer d , admit a positive integer k and matrices $A_1, \dots, A_k \in \mathcal{M}$ such that

$$\hat{\rho}(\mathcal{M}) = \rho(A_k \cdots A_1)^{1/k}?$$

The entries must be nonnegative rationals; unrestricted real entries change the status. This asks whether a finite product realizes the asymptotic growth rate.

References and status evidence

Jungers and Blondel, [On the finiteness property for rational matrices](#), Linear Algebra Appl. 428 (2008), 2283–2295, [DOI](#), abstract and reduction theorems; Jungers, *The Joint Spectral Radius: Theory and Applications*, [Chapter 4](#), pp. 63–74 (2009). They reduce this question to pairs of binary matrices in arbitrary dimension. Mejsstrik, [The finiteness conjecture for 3×3 binary matrices](#), §3 and Theorem 3.1, *Dolomites Research Notes on Approximation* 15 (2022), 24–38 (deposited on arXiv in 2025), treats the all-dimensions problem as open while proving the binary-pair case in dimension three. Searches for rational finiteness and binary-pair counterexamples through the check date found no resolution of the general question. Equivalent binary formulations count once.

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Rechecked the [published Mejsstrik paper](#), §3: binary pairs through dimension three are settled, while arbitrary dimensions remain open. Corrected its publication date to 2022; the [2025 arXiv date](#) is a deposit date. Rational-finiteness and later binary-pair searches found no general resolution.